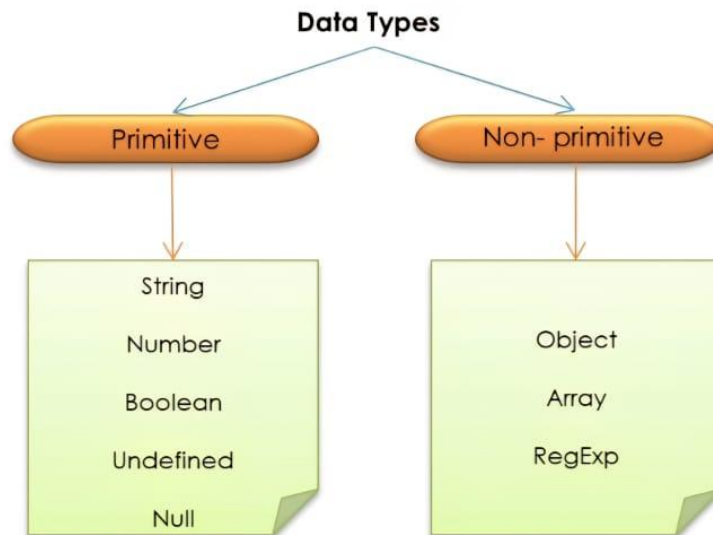


JavaScript

Data Types in JavaScript :-

In JavaScript, data types are divided into two main categories:



➤ Primitive Data Types (Immutable) -

These are the basic building blocks:

1) **String** - Represents text.

```
let a="Fortune Cloud";
console.log(a);
console.log(typeof a);
```

String Methods –

1. **charCodeAt(index)** - Returns Unicode value of character at given index.

```
let a="Fortune Cloud";
let b=a.charCodeAt(4)
console.log(b);
```

2. **toLocaleLowerCase()** - Converts string to lowercase.

```
let d=a.toLocaleLowerCase()
console.log(d);
```

3. **toLocaleUpperCase()** - Converts string to uppercase.

```
let e=a.toLocaleUpperCase()
```

```
console.log(e);
```

4. **startsWith("F")** - Checks if string starts with given substring.

```
let f=a.startsWith("F")  
console.log(f);
```

5. **endsWith("u")** - Checks if string ends with given substring.

```
let g=a.endsWith("u")  
console.log(g);
```

6. **concat()** - Joins two strings.

```
let h=a.concat("Pune")  
console.log(h);
```

7. **includes("z")** - Checks if substring exists.

```
let i=a.includes("z")  
console.log(i);
```

8. **repeat(n)** - Repeats string n times.

```
let j=a.repeat(10)  
console.log(j);
```

9. **replace(old, new)** - Replaces first match with new text.

```
let k=a.replace("Fortune","Cravita")  
console.log(k);
```

10. **slice(startIndex)** - Extracts part of string from index.

```
let l=a.slice(2)  
console.log(l);
```

2) **Number** - Represents integers, decimals, and special values (NaN, Infinity).

```
let num=34.45;  
console.log(num);  
console.log(typeof num);
```

3) **Boolean** - Represents true or false.

```
let bool=true;
console.log(bool);
console.log(typeof bool);
```

4) **Undefined** - A variable declared but not assigned a value.

```
let def;
console.log(def);
console.log(typeof def);
```

5) **Null** - Represents an intentional "empty" or "nothing".

```
let n=null;
console.log(n);
console.log(typeof n);
```

➤ **Non-Primitive Data Types (Mutable) -**

Stored as references (objects in memory):

1. **Object** - Key-value pairs.

```
let emp={
id:101,
name:"Sagar",
profile:"Software Developer",
city:"Pune"
}
console.log(emp);
console.table(emp);
console.log(Object.keys(emp));
console.log(Object.values(emp));
```

2. **Array** - Ordered collection of values.

```
let arr=[1,4,"hii",true];
console.log(arr);
console.log(typeof arr);
```

3. **Function** - A block of reusable code.

```
let b=30;           //Global Variable
function hello(){
let a=10;           //Local Variable
console.log("Hello I am a Function");
}
hello()
```

Types of Function :-

1. **Named Function** – function that has a specific name when it is defined.

```
Named Function
function add(x,y){
  console.log(x+y);
}
add(3,6)
```

2. **Anonymous Function** - A function without a name.

```
function (a,b){
  console.log(a+b);
}
(3,6)
```

3. **IIFE (Immediately Invoked Function Expression)** - A function that executes immediately after it is defined.

```
(function (a,b){
  console.log(a+b);
}
(3,6))
```

4. **Function with Expression** – It means defining a function and assigning it to a variable.

```
let mul=function(p,q){
  console.log(p*q);
}
```

5. **Arrow Function** - A shorter way to write functions using =>.

```
let mul=((p,q)=>{
  console.log(p*q);
})
mul(5,5)
```

6. **Nested Function** – A function inside another function.

```
function grandparent(){
  console.log("I am Grandparent");

  function parent(){
    console.log("I am Parent");
  }
}
```

```
function child(){  
  console.log("I am Child");  
  
  }  
  return child  
  
}  
return parent  
}  
grandparent()()()
```